

Efficacy of Dexamethasone in Adhesive Small Bowel Obstruction.

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ABSTRACT

Background

Small bowel obstruction (SBO) is a common long - term complication following abdominal surgery and a frequent cause of inpatient admissions. The most common cause of SBO is adhesions, followed by hernias and malignancy. Non - operative management strategies include bowel rest, nasogastric decompression, and Gastrografin administration, each with their own advantages and risks. Intravenous (IV) corticosteroids (Dexamethasone) have demonstrated benefit in malignant SBO but remain unexamined in adhesive SBO. The aim of this study is to evaluate the role of Dexamethasone in adhesive SBO.

Methods

A retrospective review of all patients admitted with adhesive SBO to a tertiary Colorectal Unit between 2020 and 2024 was performed. Demographics, patient results and interventional data were collected. All patients received nasogastric decompression, and IV Dexamethasone was used based on surgeon preference. For analysis, the cohort was divided into two: those who received Dexamethasone (Dex) and those who didn't (Control).

Results

A total of 105 patients were included with a mean age of 63.5 years and 54% male. There was no significant difference in baseline characteristics between the two groups. In the Dex group, 92% (46/50) resolved with non - operative management, compared to 75% (41/55) in the Control group, which was statistically significant ($p = 0.018$). There was no difference in Gastrografin use or the length of stay between the two groups.

Conclusion

IV Dexamethasone was safe and significantly reduced the need for surgical intervention in adhesive SBO. This is the first study which evaluates the use of IV Dexamethasone in non - malignant SBO and a randomised clinical trial is warranted.

Introduction

Small bowel obstruction (SBO) is a common but potentially life - threatening surgical emergency, estimated to account for 15% of hospital admissions for abdominal pain in developed countries [1, 2]. It is most commonly caused by intra - abdominal adhesions, followed by hernias, malignancy, and a further list of luminal, mural, and extraluminal pathologies [3]. Adhesive SBO is an inevitable long - term risk of any intra - abdominal surgery, with studies reporting an incidence of up to 10%, particularly in those who have undergone open procedures [4, 5, 6].

Adhesive SBO represents a significant challenge, with 20% of patients requiring surgical intervention after failure of non - operative/conservative management [7]. Paradoxically, operative management also has the potential to cause further adhesions. Current conservative treatment includes bowel rest, nasogastric decompression, and intravenous (IV) fluid resuscitation [2, 8]. While conservative strategies are often successful, these may be insufficient for patients with pronounced bowel wall edema or inflammation and may benefit from adjunct therapies. Gastrografin, a water - soluble contrast medium, has been shown to have both diagnostic and therapeutic effects in SBO [9]. Its use reduces the time to symptom resolution and length of hospital stay in SBO [10, 11]. However, it carries the risk of life - threatening aspiration

pneumonitis especially in the elderly [12].

Corticosteroids are steroid hormones with anti-inflammatory properties that can reduce oedema. Their main mechanism of action is through the suppression of the immune system via a variety of pathways, including binding to their glucocorticoid intracellular receptor which then translocates to the nucleus directly impacting transcription/expression of genes involved in the inflammatory response [13]. Their use has been well-studied in the setting of malignant bowel obstruction as safe and effective at improving symptom resolution [14, 15, 16]. We hypothesise that corticosteroids, through their anti-inflammatory and anti-oedematous effects, may offer a safe adjunctive therapeutic option in adhesive SBO. Liew et al. recently introduced steroids as a potential management option in adhesive small bowel obstruction and proposed that prospective studies followed by randomised controlled studies are essential to address this further. However, to date, there have been no studies which evaluate their use in this context [17]. The aim of this study is to investigate the outcome of corticosteroids in the management of adhesive SBO.

Methods

Study Design

A retrospective study was performed in a tertiary Colorectal centre in Sydney, Australia from 2020 to 2024. The study was reported in compliance with the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) guidelines [18]. The research protocol obtained ethics approval from the local health district committee.

Participants

All consecutive adults (≥ 18 years) diagnosed with adhesive SBO based on computed tomography of the abdomen were included. Exclusion criteria were bowel obstructions secondary to malignancy or inflammatory bowel disease, patients with peritonism or impending perforation who required immediate surgery (including obstructed hernias), or patients currently using corticosteroids. All patients received standard non-operative management including nasogastric tube decompression, bowel rest, and IV fluid and electrolyte resuscitation. If there was no resolution of obstruction after 3 days, or patients developed signs of peritonism or deterioration, surgery was indicated.

IV Dexamethasone was administered based on surgeon preference—a dosage of 8 mg on admission, followed by 4 mg twice daily on Day 2 of admission. For analysis, the cohort was divided into two: those who received IV Dexamethasone (Dex) during admission and those who did not (Control). Consent from each patient could not be gained as it is impractical to obtain consent from each patient with SBO within the last 5 years. The opportunity to gain consent no longer exists in this situation. Furthermore, de-identified data were used in the study analyses, thus research publications causing minimal harm to patients.

Variables and Outcomes

Data collected included demographics such as age, gender, and comorbidities. Other variables collected were biochemical test results on admission, hospital length of hospital, adjunct treatments such as Gastrografin, need for operative intervention, and post-operative outcomes. The primary outcome was the resolution rate without surgery. Secondary outcomes were hospital length of stay and Gastrografin use. Data were analysed and re-checked by two authors to minimise bias.

Statistical Methods

Statistical analysis was done with SPSS version 24 (IBM, USA). Mean with corresponding standard deviation for normal distribution data or median with corresponding range for non-normal distribution data were determined as appropriate. Univariate and multivariate analysis was carried out for continuous and categorical variables. The significance value was $p < 0.05$.

Results

Participants

A total of 105 patients had adhesive SBO and were included in the study. Mean age of the entire cohort was 63.5 years and 54% were males. Control group was 55 patients (52%) while Dex group was 50 patients (48%). Demographics are described in Table 1, and there were no significant differences in demographics or initial biochemical tests on admission.

Patient characteristics between the two groups.

Abbreviation: SD, standard deviation of the mean.

Results on admission.

Descriptive Data

Non operative management was successful in 92% (46/50) in the Dex group, compared to 75% (41/55) in the Control group, which was statistically significant ($p = 0.018$). All surgeries were conducted laparoscopically in the Dex group, while 64% were laparoscopically done in the Control group. As listed in Table 2, there was no difference in Gastrografin use or hospital length of stay between the two groups. No deaths occurred. Out of the four patients who received Gastrografin within the Dex group, one went on to need surgery while three resolved without surgery.

Primary and secondary outcomes.

Median value.

Statistical significance $p < 0.05$.

Discussion

Mechanical SBO causes bowel distension with enteric fluid and gas, which not only causes cardinal symptoms of abdominal distension and vomiting, but also the typical radiological findings of distended loops of bowel with air - fluid levels [19]. Bowel wall distension causes impaired venous outflow resulting in wall oedema and inflammation, which can further narrow the bowel lumen and predisposes to necrosis and perforation with peritonitis if not appropriately treated [20]. Standard conservative management of SBO comprises serial examination with bowel rest, nasogastric decompression and administration of IV fluid and electrolytes [2]. In our study, non - operative management had an overall success rate of 82%. Most significantly, our novel findings showed that IV Dexamethasone significantly reduced the need for surgical intervention in adhesive SBO.

Although the underlying mechanisms of malignant and adhesive SBO differ, both share a final common pathway of bowel wall oedema, providing a physiological rationale for corticosteroid use in symptom management. In malignant SBO, corticosteroids (typically Dexamethasone) have been shown to reduce peri - tumoral oedema, promote symptom resolution, and improve quality of life, with meta - analyses supporting their efficacy and low incidence of adverse effects [16]. Importantly, studies have highlighted their role in rapid symptom control, facilitating earlier discharge and reduced burden on healthcare resources [4]. In this context, our study is the first to examine corticosteroid use in adhesive SBO, and the findings support our hypothesis that patients who had Dexamethasone had a reduced need for surgical intervention. There is, therefore, a potential role for corticosteroids in adhesive SBO in improving non - operative outcomes and reducing healthcare costs by avoiding costly operating theatre time and equipment. It is of note that there was no significant difference in length of stay between the two groups.

Gastrografin is a water - soluble radiological contrast medium that is also used as a therapeutic adjunct in SBO. It is a hypertonic solution that draws fluids into the bowel lumen, reducing wall oedema and stimulating peristalsis [21]. Our observed rates of Gastrografin use, while similar between groups, were lower than those in other studies [22]. While this was left up to surgeon preference, we suspect this may be partly due to the risks associated with its use, including aspiration pneumonia [12]. As a result, the observed results of Dexamethasone administration are unlikely to have been significantly confounded by Gastrografin effects. However, this will be incorporated in future studies to eliminate any confounding

contribution.

We acknowledge the limitations of the present study including the retrospective design which lacks randomization and the modest sample size [23]. However, this observational cohort design included a broad patient population which still allows generalizability of findings. Importantly, our study addresses the current evidence gap and establishes a potential foundation for corticosteroids as an adjunct therapy for adhesive SBO. Based on these findings, our unit will undertake a prospective and a multi - centre randomized controlled trial to provide high quality evidence, establish a corticosteroid protocol and also explore short - term and long - term outcomes such as SBO recurrence rates.

Conclusion

The administration of IV Dexamethasone in adhesive SBO significantly reduced the need for surgical management. This is the first study which evaluates IV Dexamethasone in adhesive obstruction and warrants further randomised clinical trials.

Author Contributions

Supun M. Bakmiwewa: writing – original draft, methodology, writing – review and editing. Mina Sarofim: conceptualization, methodology, writing – review and editing, formal analysis. Ernest Cheng: methodology, writing – review and editing. Dean Yeh: conceptualization, writing – review and editing. Scott MacKenzie: conceptualization, writing – review and editing. Katherine Gibson: conceptualization, writing – review and editing. Andrew Gilmore: conceptualization, writing – review and editing, supervision. Assad Zahid: conceptualization, writing – review and editing, supervision.

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Disclosure

The authors have nothing to report.

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Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

References

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